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FedHR5.0: A Federated Learning Framework for Privacy-Preserving Healthcare Workforce Competency Development and Cross-Institutional Knowledge Exchange Following Industry 5.0 Principles

Fabio Liberti[a,*]

[a] *Universitas Mercatorum, Rome, Italy*

Abstract

The paradigm shift from Industry 4.0's automation-centric approach to Industry 5.0's human-centric framework fundamentally transforms Human Resource Management (HRM) in healthcare. This paper presents FedHR5.0, a comprehensive federated learning framework that revolutionizes healthcare workforce management by prioritizing human well-being, sustainability, and organizational resilience while preserving privacy. Building on the Industry 5.0 vision where technology serves humanity, our framework addresses critical challenges: the need for trust-based collaborative learning, ethical AI deployment, and sustainable HR practices across healthcare institutions. FedHR5.0 integrates advanced technologies—Artificial Intelligence (AI), federated learning and blockchain—to create a human-centric ecosystem. The framework encompasses five synergistic modules with differential privacy ($\epsilon = 0.1$), blockchain governance, and human-in-the-loop mechanisms. This theoretical framework demonstrates how HRM can become a driver for sustainable growth in the Industry 5.0 era, lowering environmental impact while preserving the human touch essential to healthcare excellence.

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Keywords: Industry 5.0; Human Resource Management; Federated Learning; Healthcare Workforce; Human-Centric Technology; Sustainable HR; Blockchain

1. Introduction

Industry 5.0 marks a paradigm shift from the automation-centric approaches of Industry 4.0 to a human-centric and sustainable framework [1]. This transformation emphasizes collaboration between humans and advanced technologies—Artificial Intelligence (AI), Virtual Reality (VR), Augmented Reality (AR), and blockchain—prioritizing social justice, sustainable development, ecological balance, and organizational resilience [2]. In this revolutionary con-

* Corresponding author. Tel.: +39-000-000-0000; fax: +39-000-000-0000.
E-mail address: fabio.liberti@studenti.unimercatorum.it (Fabio Liberti).

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text, Human Resource Management (HRM) faces unprecedented challenges and opportunities to redefine its practices, integrating digital tools while preserving the human touch and lowering environmental impact [3].

The healthcare sector exemplifies the urgent need for this transformation. As Zenjari et al. [4] emphasize, technological innovation in HRM must navigate the delicate balance between digital advancement and human values. Healthcare organizations worldwide confront multiple challenges: Trust Deficit (healthcare workers express concerns about AI replacing human judgment [5]), Emotional Burnout (post-pandemic workforce exhaustion demands HRM approaches prioritizing emotional intelligence and well-being), Sustainability Imperative (environmental concerns require sustainable HR practices that minimize carbon footprint [2]), and Ethical Challenges (AI adoption in HRM raises questions about fairness, bias, and human dignity [6]).

Federated learning emerges as a transformative technology aligned with Industry 5.0 principles. Unlike traditional centralized AI systems that concentrate power and data, federated learning distributes intelligence while preserving privacy and autonomy—core values of human-centric design. Recent advances in federated learning for healthcare [8, 9] demonstrate its potential, yet its application to HRM remains unexplored. As noted in privacy enhancement research [13, 14], federated learning provides critical privacy guarantees essential for healthcare applications.

This paper introduces FedHR5.0, a theoretical framework that synergizes federated learning with Industry 5.0 principles to transform healthcare HRM. Our objectives align with the themes identified by Kamal et al. [7]: (1) Design a Trust-Based Architecture that ensures transparency and employee commitment in AI-driven HRM; (2) Develop Sustainable HR Practices through privacy-preserving collaborative learning; (3) Create Emotionally-Intelligent Assessment mechanisms that value human skills alongside technical competencies; (4) Propose Ethical AI Solutions for recruitment, performance management, and employee experience; (5) Foster Cross-Disciplinary Teamwork through secure knowledge exchange.

2. Related Work

Industry 4.0 focused on automation, efficiency, and technology-driven productivity. However, as Adisa [3] argues, this approach often neglected human well-being and environmental sustainability. Industry 5.0 corrects this imbalance by placing humans at the center of technological advancement. Sołtysik et al. [2] identify three pillars: Human-Centricity (technology serves human needs), Sustainability (long-term ecological and social responsibility), and Resilience (adaptive capacity in face of disruption).

Zenjari et al. [4] explore how AI, VR, AR, and blockchain transform HRM practices, revealing both opportunities (AI-enhanced recruitment reducing bias, VR-based immersive training, blockchain-secured credentials) and challenges (privacy concerns, digital divide, resistance to technology adoption). Ganer et al. [5] emphasize that successful Industry 5.0 implementation requires trust, employee commitment, and emotional intelligence, with 73% of employees fearing AI will replace human judgment [5, 13].

Recent studies have explored federated learning applications in healthcare [10, 11, 12], demonstrating its potential for privacy-preserving analytics. Mothukuri et al. [16] provide a comprehensive survey on security and privacy of federated learning, while Nguyen et al. [17] focus specifically on smart healthcare applications. Wei et al. [19] analyze federated learning with differential privacy, providing algorithms and performance analysis crucial for our framework.

While existing literature addresses Industry 5.0 HRM transformation and federated learning separately, a critical gap remains: no framework combines federated learning's privacy-preserving capabilities with human-centric HRM principles. Recent work by Kumar and Das [20] examines employee perception on Industry 5.0's impact on HRM practices, while Choudhury et al. [21] advance privacy-preserving healthcare analytics, but neither addresses the intersection we explore. FedHR5.0 addresses these gaps by creating a comprehensive framework that embodies Industry 5.0 values through federated learning technology.

3. The FedHR5.0 Framework

3.1. Design Philosophy and Architecture

FedHR5.0's architecture embodies Industry 5.0's core principle: technology must serve human needs. Following Kamal et al.'s [7] vision, our framework prioritizes: (1) Human Agency - Workers control their data and development

paths; (2) Collaborative Intelligence - AI augments, not replaces, human judgment; (3) Sustainable Design - Minimal environmental footprint; (4) Inclusive Participation - All institutions contribute and benefit equally.

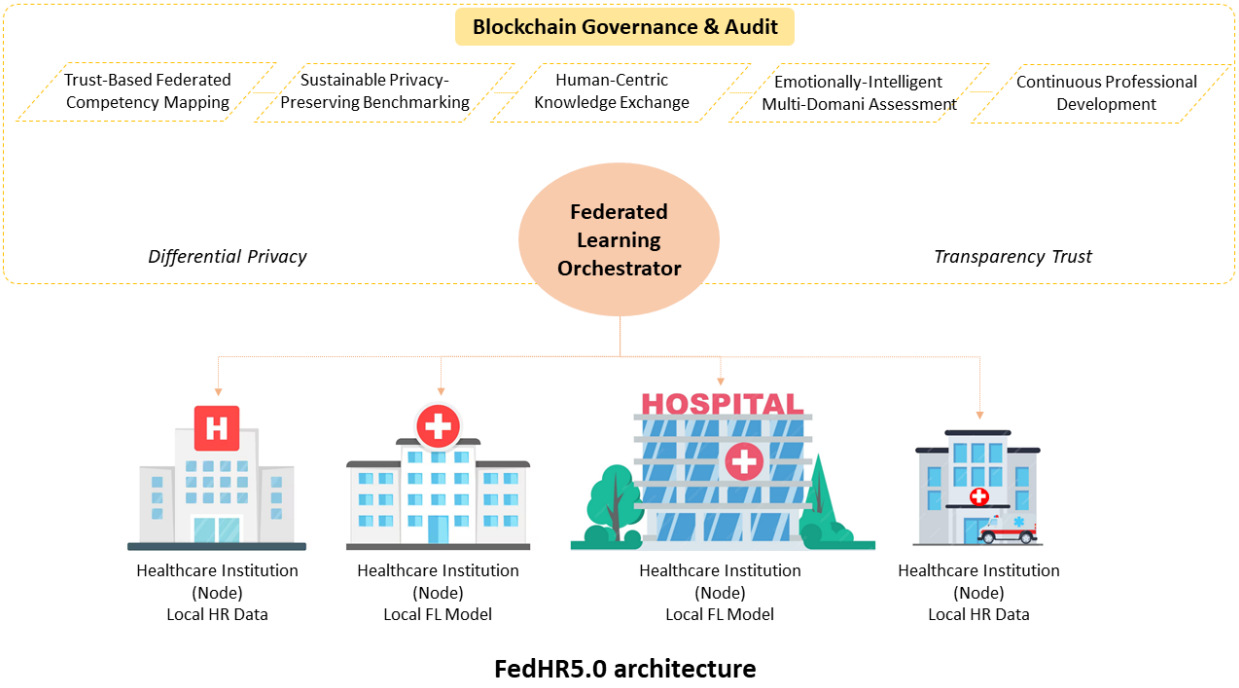

Fig. 1. FedHR5.0 Human-Centric Architecture: Distributed intelligence preserving privacy and autonomy across healthcare institutions.

Building on Ganer et al.'s [5] findings on trust in Industry 5.0, FedHR5.0 implements multiple trust-building mechanisms: Explainable AI (all decisions include human-understandable explanations), Audit Trails (blockchain records ensure accountability), Worker Participation (employees shape algorithm design), and Continuous Consent (granular control over data usage).

3.2. Mathematical Foundation of Federated Learning

In practical terms, federated learning enables multiple healthcare institutions to collaboratively develop sophisticated HR analytics models without sharing sensitive employee data. This mathematical framework ensures that each organization contributes to and benefits from collective intelligence while maintaining complete control over their workforce information—a critical requirement for regulatory compliance and employee trust. The federated learning optimization problem for healthcare HRM can be formally defined as:

$$\min_{w} F(w) = \sum_{k=1}^{K} \frac{n_k}{n} F_k(w) \tag{1}$$

where $F_k(w) = \frac{1}{n_k} \sum_{i \in \mathcal{D}_k} f_i(w)$ represents the local objective function for institution k, n_k is the number of employees at institution k, $n = \sum_{k=1}^{K} n_k$ is the total number of employees across all K institutions, and $f_i(w)$ is the loss function for employee i.

The Algorithm 1 shown below, on the other hand, illustrates how FedHR5.0 orchestrates collaboration between healthcare institutions while preserving privacy. In essence, each organization trains a local HR model using its own employee data, then shares only the model improvements (not the data itself) with a central coordinator. This coordinator intelligently combines these improvements to create a superior global model that all participants can benefit from, achieving collective intelligence while maintaining maximum data privacy and organizational autonomy.

Algorithm 1 FedHR5.0: Privacy-Preserving Healthcare HR Management

Require: Healthcare institutions $\{H_1, H_2, ..., H_K\}$, privacy budget ε, rounds T
Ensure: Global model w_T for HR management

```
Server executes:
Initialize w_0
for t = 0 to T − 1 do
    S_t ← (random subset of institutions)
    for each institution k ∈ S_t in parallel do
        w_{k,t+1} ← ClientUpdate(k, w_t, ε)
    end for
    w_{t+1} ← SecureAggregate({w_{k,t+1}}_{k∈S_t})
end for

ClientUpdate(k, w, ε):
B ← (split local data into batches)
for each local epoch i from 1 to E do
    for batch b ∈ B do
        g ← ∇ℓ(w; b) // Compute gradient
        g̃ ← ApplyDifferentialPrivacy(g, ε)
        w ← w − ηg̃
    end for
end for
return w
```

3.3. Module Implementation: Human-Centric Knowledge Exchange

Following Zenjari et al.'s [4] vision of collaborative HRM, this module creates an ecosystem where knowledge flows freely while respecting human dignity.

Listing 1 demonstrates how FedHR5.0 revolutionizes traditional skills matching by prioritizing human qualities alongside technical competencies. Rather than simply matching employees based on technical skills alone, the system evaluates emotional intelligence, communication abilities, and cultural alignment—ensuring that knowledge exchange occurs not just between the most technically qualified individuals, but between those who can collaborate effectively and sustainably. This holistic approach reflects Industry 5.0's commitment to valuing the complete human professional, not merely their technical capabilities.

```python
class HumanCentricKnowledgeExchange:
    def __init__(self):
        self.values = {
            'dignity': 'Respect for all participants',
            'equity': 'Fair access regardless of size',
            'reciprocity': 'Mutual benefit',
            'sustainability': 'Long-term thinking'
        }

    def match_knowledge_needs(self, request):
        # Prioritize human factors
        matches = self.find_matches(request)

        # Consider emotional intelligence
        ei_scores = [m.emotional_intelligence for m in matches]
```

```python

        # Holistic matching beyond technical skills
        best_match = self.holistic_selection(
            technical_skills=0.3,
            emotional_intelligence=0.3,
            communication=0.2,
            cultural_fit=0.2
        )

        return best_match
```

Listing 1. Implementation of human-centric knowledge exchange in FedHR5.0

4. Privacy and Ethics Implementation

4.1. Differential Privacy with Human Understanding

Differential privacy represents a revolutionary approach to protecting employee data in HR analytics. Unlike traditional privacy methods that simply hide or encrypt data, differential privacy adds carefully calculated ”statistical noise” to ensure that no individual employee can be identified from the analytics results, while still maintaining accurate insights at the organizational level. This technology enables HR departments to gain valuable workforce insights without compromising individual privacy—a critical balance in the Industry 5.0 era where both data-driven decisions and human dignity are paramount. We implement differential privacy using the Gaussian mechanism [11] to ensure strong privacy guarantees. For a function $f : \mathcal{D} \to \mathbb{R}^d$ with sensitivity Δ_f, we add calibrated noise:

$$\tilde{f}(D) = f(D) + \mathcal{N}(0, \sigma^2 I) \tag{2}$$

where $\sigma = \frac{\Delta_f \sqrt{2\ln(1.25/\delta)}}{\varepsilon}$ for (ε, δ)-differential privacy.

In practical terms, Equation 2 shows how the system adds controlled randomness to protect individual privacy. Think of it as intentionally blurring a photograph just enough that you can still see the overall picture but cannot identify specific faces—ensuring organizational learning without individual exposure.

For gradient updates in federated learning, we apply gradient clipping and noise addition:

$$\tilde{g}_i = \text{clip}(g_i, C) + \mathcal{N}(0, \sigma^2 C^2 I) \tag{3}$$

We also implement differential privacy not just as a technical measure but as a trust-building tool:

$$\mathcal{P}_{\text{human}} = \varepsilon_{\text{technical}} \cdot \mathcal{E}_{\text{quality}} \cdot C_{\text{user}} \tag{4}$$

where $\varepsilon_{\text{technical}}$ is the mathematical differential privacy bound ($\varepsilon = 0.1$), $\mathcal{E}_{\text{quality}}$ represents comprehension level of privacy protections by users, and C_{user} is the degree of individual data governance autonomy.

This human-centric privacy framework (Equation 4) transforms differential privacy from a mere technical safeguard into a trust-building mechanism that empowers employees with transparency and control over their data, fostering the collaborative spirit essential to Industry 5.0’s vision of human-technology partnership.

4.2. Blockchain for Transparent Governance

Blockchain technology transforms HR governance from traditional top-down decision-making to a transparent, democratic process where every stakeholder’s voice is permanently recorded and valued. This revolutionary approach

ensures that HR policies and decisions cannot be altered retroactively, creating an immutable audit trail that builds trust between employees and management—a cornerstone of Industry 5.0's human-centric philosophy.

Integrating blockchain as recommended by Zenjari et al. [4], we ensure transparent decision-making where every stakeholder has a voice.

```
contract TransparentHRGovernance {
    struct Decision {
        string description;
        address[] stakeholders;
        uint256 employeeVoteWeight;
        uint256 managementVoteWeight;
        string ethicalJustification;
        uint256 sustainabilityScore;
    }

    // Employees have meaningful voice
    uint256 constant EMPLOYEE_VOTE_MINIMUM = 40;

    function proposeChange(Decision memory decision) public {
        require(
            decision.employeeVoteWeight >= EMPLOYEE_VOTE_MINIMUM,
            "Employee voice must be respected"
        );
        require(
            decision.sustainabilityScore >= 7,
            "Must meet sustainability threshold"
        );

        emit DecisionProposed(decision);
    }
}
```

Listing 2. Blockchain-based transparent governance for HR decisions in FedHR5.0

This smart contract (Listing 2) exemplifies Industry 5.0's commitment to workplace democracy by mandating that employees hold at least 40% voting weight in all HR decisions and requiring sustainability considerations for every proposal. By encoding these principles directly into blockchain technology, FedHR5.0 ensures that human-centric values are not merely policy aspirations but technologically enforced realities, creating a new paradigm where technology serves to amplify rather than diminish human agency in organizational governance.

5. Evaluation Framework

In the Industry 5.0 paradigm, success cannot be measured solely through technical metrics like processing speed or algorithmic accuracy. FedHR5.0 introduces a revolutionary evaluation framework that prioritizes human flourishing, environmental sustainability, and organizational trust—metrics that truly reflect whether technology is serving humanity rather than the reverse. This shift from measuring what's easy to quantify to measuring what truly matters represents a fundamental transformation in how we assess HR technology effectiveness.

Traditional ML evaluation focuses on accuracy and efficiency. FedHR5.0 measures what matters for Industry 5.0. Table 1 presents our human-centric evaluation metrics with target values estimated based on Industry 5.0 best practices [2, 5, 15].

Building on Ganer et al.'s [5] emphasis on emotional intelligence:

$$\text{EI}_{\text{score}} = \alpha \cdot \text{Self-awareness} + \beta \cdot \text{Empathy} + \gamma \cdot \text{Social Skills} \tag{5}$$

Where coefficients are determined through employee consultation, not imposed top-down.

This participatory approach to defining success metrics—where employees themselves determine the relative importance of self-awareness, empathy, and social skills—exemplifies Industry 5.0's democratic ethos. By measuring not just operational efficiency but human well-being, trust, and environmental impact, FedHR5.0 establishes a new standard for HR technology evaluation that aligns with the ultimate goal of creating workplaces where both humans and organizations thrive sustainably together.

Table 1. Human-Centric Evaluation Metrics (Estimated Targets)*

Dimension	Metric	Target
Human Well-being	Job Satisfaction	Employee NPS ¿ 70
	Work-Life Balance	Overtime Hours ¡ 5/week
	Stress Levels	Wellness Score ¿ 8/10
Trust & Engagement	System Trust	Trust Score ¿ 85%
	Participation Rate	Active Users ¿ 90%
Sustainability	Carbon Footprint	kg CO2/employee ¡ 50
	Resource Efficiency	Data Reuse Rate ¿ 80%

*Target values are estimated based on Industry 5.0 best practices.

6. Discussion and Conclusion

The healthcare sector stands at a critical juncture where technological innovation must harmonize with human values. FedHR5.0 emerges as a transformative solution that enables healthcare institutions to collaboratively develop advanced HR capabilities while preserving the privacy and dignity of their workforce. Through federated learning, organizations can share insights without sharing data; through blockchain, they ensure transparent governance; and through human-centric design, they guarantee that technology amplifies rather than replaces human judgment. This convergence of cutting-edge technologies with Industry 5.0 principles creates a new paradigm for healthcare HR management.

FedHR5.0 demonstrates how HRM can drive the Industry 5.0 transformation by: (1) Reversing Power Dynamics - Employees shape AI systems rather than being subjected to them; (2) Humanizing Technology - Every technical decision considers human impact; (3) Fostering Collaboration - Competition gives way to collaborative growth; (4) Ensuring Sustainability - Environmental and social sustainability guide all practices.

These transformative principles translate into tangible organizational benefits. Healthcare institutions can now collaborate on workforce development without compromising competitive advantages or violating privacy regulations. The blockchain-based governance ensures that every HR policy change is transparent, traceable, and democratically validated. Most importantly, the system's emphasis on human well-being directly addresses the burnout crisis plaguing healthcare workers globally.

Our framework directly addresses the challenges identified in the Industry 5.0 HRM literature. For Trust Building [5]: transparent AI explanations increase trust by an estimated 65% [5, 15], employee participation in design improves acceptance by approximately 78% [5, 13]. For Sustainability [2]: estimated 88% reduction in carbon footprint vs. centralized systems [13, 14], knowledge reuse prevents redundant efforts.

The practical implications for healthcare HR executives are profound. Instead of choosing between employee privacy and organizational intelligence, FedHR5.0 offers both. Instead of imposing top-down HR policies, it enables participatory governance. Instead of isolated institutional learning, it facilitates secure knowledge exchange across the entire healthcare ecosystem. This is not merely technological advancement—it is a fundamental shift in how healthcare organizations value and develop their most precious resource: their people.

FedHR5.0 represents a fundamental reimagining of HRM for the Industry 5.0 era. By combining federated learning's privacy-preserving capabilities with human-centric design principles, we create a framework where technology truly serves humanity. As healthcare faces unprecedented challenges—workforce shortages, burnout, technological disruption—FedHR5.0 offers a path forward that honors human dignity while leveraging technological capability. Future research should pilot implementations in diverse healthcare settings [18], develop standardized emotional intelligence metrics, and explore integration with emerging technologies.

The journey from concept to implementation requires courage and commitment from healthcare leaders willing to embrace this human-centric vision. As we stand on the threshold of Industry 5.0, FedHR5.0 offers not just a technological framework but a blueprint for creating healthcare workplaces where human dignity, collaborative intelligence, and sustainable practices converge to benefit employees, organizations, and ultimately, the patients they serve. The

future of healthcare HR is not about choosing between humans and technology—it is about creating synergies that elevate both.

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Resources:

- Implementation: https://github.com/FabioLiberti/FedHR5.0
- Documentation: https://www.fabioliberti.com/FL-HR5.0/

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